



Association between postoperative hypotension and acute kidney injury after noncardiac surgery: a historical cohort analysis

Association entre hypotension postopératoire et insuffisance rénale aiguë après une chirurgie non cardiaque : une analyse de cohorte historique

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Abstract

Purpose *The extent to which postoperative hypotension contributes to renal injury remains unclear, much less what the harm thresholds might be. We therefore tested the primary hypothesis that there is an absolute hypotensive arterial pressure threshold for acute kidney injury during the initial seven days after noncardiac surgery.*

Methods *We conducted a single-centre historical cohort analysis of adults who had noncardiac surgery and had creatinine recorded preoperatively and postoperatively. Our exposure was the lowest postoperative mean arterial pressure, defined as the average of the three lowest postoperative pressure measurements. Our primary analysis was the association between the lowest mean arterial pressure and acute kidney injury, defined according to Kidney Disease: Improving Global Outcomes initiative criteria. Our analysis was adjusted for potentially relevant confounding factors including intraoperative hypotension.*

Results *Among 64,349 patients analyzed, 2,812 (4.4%) patients had postoperative acute kidney injury.*

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Each 5-mm Hg decrease in the lowest mean arterial pressure was associated with a 28% (97.5% confidence interval [CI], 23 to 32; $P < 0.001$) increase in the odds of acute kidney injury for lowest mean arterial pressures < 80 mm Hg. Higher lowest pressures were not associated with acute kidney injury (odds ratio, 1.08; 97.5% CI, 0.99 to 1.17; $P = 0.04$) for each 5-mm Hg decrease in the lowest mean arterial pressure.

Conclusion Postoperative hypotension, defined as the lowest postoperative mean arterial pressure < 80 mm Hg, was associated with acute kidney injury after noncardiac surgery. A prospective trial will be required to determine whether the observed association is causal and thus amenable to modification.

Résumé

Objectif Nous ne savons pas dans quelle mesure l'hypotension postopératoire contribue aux lésions rénales, et nous connaissons encore moins les seuils de lésion. Nous avons donc testé l'hypothèse primaire selon laquelle il existerait un seuil absolu de tension artérielle hypotensive pour l'insuffisance rénale aiguë au cours des sept premiers jours suivant une chirurgie non cardiaque.

Méthode Nous avons mené une analyse de cohorte historique monocentrique auprès d'adultes ayant bénéficié d'une chirurgie non cardiaque et pour lesquels les taux de créatinine avant et après l'opération avaient été enregistrés. Notre exposition était la tension artérielle moyenne postopératoire la plus basse, définie comme la moyenne des trois mesures de tension postopératoire les plus basses. Notre analyse principale a porté sur l'association entre la tension artérielle moyenne la plus basse et l'insuffisance rénale aiguë, définies selon les critères de l'initiative KDIGO (Kidney Disease: Improving Global Outcomes). Notre analyse a été ajustée pour tenir compte des facteurs de confusion potentiellement pertinents, notamment de l'hypotension peropératoire.

Résultats Parmi les 64 349 patient-es analysé-es, 2812 (4,4 %) ont présenté une insuffisance rénale aiguë postopératoire. Chaque diminution de 5 mm Hg de la tension artérielle moyenne la plus faible était associée à une augmentation de 28 % (intervalle de confiance [IC] de 97,5 %, 23 à 32; $P < 0,001$) des risques d'insuffisance rénale aiguë pour les tensions artérielles moyennes les plus faibles < 80 mm Hg. Des tensions les plus faibles plus hautes n'ont pas été associées à une insuffisance rénale aiguë (rapport de cotes, 1,08; IC 97,5 %, 0,99 à 1,17; $P = 0,04$) pour chaque diminution de 5 mm Hg de la tension artérielle moyenne la plus faible.

Conclusion L'hypotension postopératoire, définie comme la tension artérielle moyenne postopératoire < 80 mm Hg la plus basse, a été associée à une insuffisance rénale aiguë après une chirurgie non cardiaque. Une étude prospective

sera nécessaire pour déterminer si l'association observée est causale et donc susceptible d'être modifiée.

Keywords acute kidney injury · anesthesia · hypotension · postoperative

Acute kidney injury (AKI) is common after noncardiac surgery, with a reported incidence ranging from 7 to 13%.^{1–3} Most postoperative AKI—including higher stages—resolves in the year after surgery. Nevertheless, a quarter of patients who develop AKI have the same or worse renal injury one to two years after surgery.⁴ There are currently no documented ways of preventing perioperative AKI.

Intraoperative hypotension is associated with AKI after noncardiac surgery. The duration of intraoperative pressure excursions is usually limited because anesthesiologists intervene. Postoperative blood pressure differs in that hypotension is often prolonged and profound.⁵ At some level, hypotension causes organ injury. It is thus unsurprising that hypotension is associated with myocardial injury, myocardial infarction, and mortality in patients recovering from noncardiac surgery.^{6–8} Nevertheless, the extent to which postoperative hypotension might contribute to AKI remains unclear.

We therefore tested the primary hypothesis that there is an absolute mean arterial pressure (MAP) threshold for postoperative hypotension that is associated with AKI during the initial seven days after noncardiac surgery. We also evaluated whether the association depends on baseline blood pressure; that is, whether there is an interaction with baseline MAP.

Methods

This study was approved by the Cleveland Clinic institutional review board (IRB) with waived consent (IRB#: 19-085; date of the approval: 27 March 2019; IRB executive director: Bridget Howard; Cleveland, OH, USA). This manuscript adheres to the applicable Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines. We aimed to conduct a historical cohort analysis exploring an association between the lowest and the highest postoperative MAP and AKI after noncardiac surgery.

Study population

Data were obtained from the Cleveland Clinic Perioperative Health Documentation System, an electronic medical record-based registry of noncardiac

surgery patients who had surgery at the Cleveland Clinic between 6 January 2009 and March 2019. We included noncardiac inpatients with American Society of Anesthesiologists Physical Status less than V with the following inclusion criteria: 1) creatinine concentration recorded within 30 days before surgery and within the first seven postoperative days; 2) blood pressure recorded in the preanesthesia care evaluation clinic or during other preoperative appointments within six months before surgery; 3) initial postoperative care in a routine surgical ward rather than an intensive care unit or special monitoring unit; and 4) at least five blood pressure measurements on the ward.

Patients were excluded based on the following criteria: 1) chronic kidney disease defined as preoperative estimated glomerular filtration rate of less than $60 \text{ mL} \times \text{min}^{-1} \times 1.73 \text{ m}^{-2}$ or on dialysis; 2) urologic procedures including relief of urinary obstruction, nephrectomy, and renal transplantation, as described in Electronic Supplementary Material (ESM) eTable 1; 3) preoperative creatinine exceeding $1.3 \text{ mg} \cdot \text{dL}^{-1}$; 4) anesthesia lasting less than 60 min (induction to emergence); and 5) invalid or unavailable postoperative blood pressure data for more than nine consecutive hours.

Measurements

The primary outcome was postoperative AKI, defined by the Kidney Disease: Improving Global Outcomes (KDIGO) Clinical Practice Guidelines for Acute Kidney Injury^{9,10} (ESM eTable 2). As usual in perioperative studies, urine output was ignored because these data are rarely available. Acute kidney injury diagnosis was based on the difference between the last creatinine concentration before surgery and the highest postoperative concentration measured within the first seven postoperative days or until discharge.

Our exposures of interest were the lowest and the highest MAPs after surgery, defined as the average of the three lowest and three highest postoperative MAP values within seven postoperative days, until discharge, or until AKI was diagnosed. All MAP measurements in the postanesthesia care unit (PACU) were averaged and considered as a single measurement that was then pooled with ward MAP measurements.

Statistical analysis

For reasons mentioned in the results section, we restricted our analysis to hypotension. We first fitted a multivariable logistic regression model for the exposure (the lowest MAP) with natural restricted cubic splines of the exposure at 10%, 50%, and 90% percentile location to assess the linearity of the exposure and AKI, adjusting for confounders

in Table 1. We adjusted for the intraoperative time-weighted average MAP $< 65 \text{ mm Hg}$, to compensate for intraoperative hypotension. Furthermore, we assessed the interaction between the exposure and quartiles of baseline blood pressure, which was defined as the average of all MAP recordings in the six months before surgery, excluding measurements made during a hospital stay.

Since no qualitative (i.e., in opposite directions) interaction was found between baseline blood pressure and the exposure, we removed the interaction term and proceeded to assess the marginal relationship between the exposure and AKI. Since a statistically significant nonlinear relationship between the lowest MAP and AKI was found ($P < 0.001$ for a nonlinear spline term), we added a piece-wise “linear” knot (i.e., relationships before and after the knot are forced to be linear but can have different slopes) at 80 mm Hg for the lowest MAP based on relationship visualization. This method allowed us to estimate potentially different odds ratios before and after the knot in the multivariable logistic regression model.

Based on the visualized linear knot, we additionally reported odds ratios treating the lowest MAP as categorical: the lowest MAP groups included the lowest MAP $> 80 \text{ mm Hg}$ as the reference group, the lowest MAP of $60\text{--}80 \text{ mm Hg}$, and the lowest MAP $\leq 60 \text{ mm Hg}$.

Finally, to see which stage of AKI strongly drives the association, we did a post hoc sensitivity analysis by fitting a multinomial logistic regression model, with all the stages being separated and stage 0 being considered as the reference stage, so we could estimate the odds ratio between pairwise stages associated with exposure. As shown in the results section, there were too few events in stage 3 ($N = 140$), so stages 2 and 3 were combined together as stage 2 or 3 (stage 2/3). A linear piece-wise spline term at 80 mm Hg of lowest MAP was also added like primary analysis to quantify the association differently below, above, or equal to 80 mm Hg .

The overall significance level for the primary analysis and for testing the interaction with baseline MAP were each 0.05. We used Bonferroni correction for multiple testing within the exposure. The significance level was 0.025 below and above the threshold for exposure. All analyses were conducted in SAS studio 3.7 Basic Edition (2012–2017, SAS Institute Inc., Cary, NC, USA).

Results

Study population

There were 65,936 patients who met all inclusion and exclusion criteria. Of these, 1,587 (2.4%) had missing potential confounding variables and were not included, leaving 64,349 patients in the final

Table 1 Baseline and demographic characteristics

Factor	Total N = 64,349	No AKI N = 61,537	AKI N = 2,812	P value
Age (yr), mean (SD)	56.7 (15.5)	56.4 (15.6)	62.5 (14.2)	< 0.001 ^a
Female, n/total N (%)	38,579/64,349 (60%)	36,990/61,537 (60%)	1,589/2,812 (57%)	< 0.001 ^b
Race, n/total N (%)				< 0.001 ^b
Asian	460/64,349 (0.7%)	448/61,537 (0.7%)	12/2,812 (0.4%)	
Black	7,402/64,349 (11.5%)	6,975/61,537 (11.3%)	427/2,812 (15.2%)	
White	55,274/64,349 (85.9%)	52,956/61,537 (86.1%)	2,318/2,812 (82.4%)	
Other	1,213/64,349 (1.9%)	1,158/61,537 (1.9%)	55/2,812 (2.0%)	
BMI, mean (SD)	30.6 (8.6)	30.5 (8.6)	31.7 (8.5)	< 0.001 ^a
Baseline MAP (mm Hg), mean (SD)	93.0 (9.9)	93.0 (9.9)	93.3 (10.6)	0.14 ^a
ASA Physical Status, n/total N (%)				< 0.001 ^b
I	851/64,349 (1.3%)	843/61,537 (1.4%)	8/2,812 (0.3%)	
II	19,111/64,349 (29.7%)	18,717/61,537 (30.4%)	394/2,812 (14.0%)	
III	39,989/64,349 (62.1%)	37,943/61,537 (61.7%)	2,046/2,812 (72.8%)	
IV	4,398/64,349 (6.8%)	4,034/61,537 (6.6%)	364/2,812 (12.9%)	
POA risk index, mean (SD)	−4.6 (0.71)	−4.7 (0.71)	−4.4 (0.65)	< 0.001 ^a
Angiotensin II receptor blockers, n/total N (%)	3,737/64,349 (5.8%)	3,455/61,537 (5.6%)	282/2,812 (10.0%)	< 0.001 ^b
Calcium channel blockers, n/total N (%)	6,923/64,349 (10.8%)	6,389/61,537 (10.4%)	534/2,812 (19.0%)	< 0.001 ^b
Beta blockers, n/total N (%)	16,896/64,349 (26.3%)	15,784/61,537 (25.6%)	1,112/2,812 (39.5%)	< 0.001 ^b
Diuretics, n/total N (%)	12,265/64,349 (19.1%)	11,379/61,537 (18.5%)	886/2,812 (31.5%)	< 0.001 ^b
ACEI, n/total N (%)	8,426/64,349 (13.1%)	7,811/61,537 (12.7%)	615/2,812 (21.9%)	< 0.001 ^b
Statins, n/total N (%)	13,741/64,349 (21.4%)	12,890/61,537 (20.9%)	851/2,812 (30.3%)	< 0.001 ^b
PPIs, n/total N (%)	15,351/64,349 (23.9%)	14,443/61,537 (23.5%)	908/2,812 (32.3%)	< 0.001 ^b
Congestive heart failure, n/total N (%)	2,257/64,349 (3.5%)	2,004/61,537 (3.3%)	253/2,812 (9.0%)	< 0.001 ^b
Peripheral vascular disease, n/total N (%)	2,818/64,349 (4.4%)	2,611/61,537 (4.2%)	207/2,812 (7.4%)	< 0.001 ^b
Stroke, n/total N (%)	1,226/64,349 (1.9%)	1,158/61,537 (1.9%)	68/2,812 (2.4%)	0.04 ^b
Liver disease, n/total N (%)	3,882/64,349 (6.0%)	3,727/61,537 (6.1%)	155/2,812 (5.5%)	0.24 ^b
Smoking, n/total N (%)	8,412/64,349 (13.1%)	8,104/61,537 (13.2%)	308/2,812 (11.0%)	< 0.001 ^b
Hypertension, n/total N (%)	31,439/64,349 (48.9%)	29,504/61,537 (47.9%)	1,935/2,812 (68.8%)	< 0.001 ^b
Diabetes mellitus, n/total N (%)	11,188/64,349 (17.4%)	10,435/61,537 (17.0%)	753/2,812 (26.8%)	< 0.001 ^b
Obesity, n/total N (%)	15,342/64,349 (23.8%)	14,454/61,537 (23.5%)	888/2,812 (31.6%)	< 0.001 ^b
Coronary artery disease, n/total N (%)	3,674/64,349 (5.7%)	3,375/61,537 (5.5%)	299/2,812 (10.6%)	< 0.001 ^b
Metastatic cancer, n/total N (%)	4,008/64,349 (6.2%)	3,748/61,537 (6.1%)	260/2,812 (9.2%)	< 0.001 ^b
Solid tumour without metastasis, n/total N (%)	6,464/64,349 (10.0%)	6,057/61,537 (9.8%)	407/2,812 (14.5%)	< 0.001 ^b
Number of antihypertensive agents, median [IQR]	0.00 [0.00–0.00]	0.00 [0.00–0.00]	0.00 [0.00–1.00]	< 0.001 ^c
Emergent surgery, n/total N (%)	1,311/64,349 (2.0%)	1,226/61,537 (2.0%)	85/2,812 (3.0%)	< 0.001 ^b
Surgery type, operation on the, n/total N (%)				< 0.001 ^b
Cardiovascular system	2,210/64,349 (3.4%)	2,133/61,537 (3.5%)	77/2,812 (2.7%)	
Digestive system	26,104/64,349 (40.6%)	24,798/61,537 (40.3%)	1,306/2,812 (46.4%)	
Endocrine system	1,120/64,349 (1.7%)	1,093/61,537 (1.8%)	27/2,812 (1.0%)	
Female genital organs	6,354/64,349 (9.9%)	6,145/61,537 (10.0%)	209/2,812 (7.4%)	
Hemic and lymphatic system	828/64,349 (1.3%)	798/61,537 (1.3%)	30/2,812 (1.1%)	
Integumentary system	2,279/64,349 (3.5%)	2,171/61,537 (3.5%)	108/2,812 (3.8%)	
Male genital organs	686/64,349 (1.1%)	662/61,537 (1.1%)	24/2,812 (0.9%)	
Musculoskeletal	14,831/64,349 (23.0%)	14,110/61,537 (22.9%)	721/2,812 (25.6%)	
Nervous system	6,200/64,349 (9.6%)	6,081/61,537 (9.9%)	119/2,812 (4.2%)	
Respiratory system	461/64,349 (0.7%)	442/61,537 (0.7%)	19/2,812 (0.7%)	
Urinary system	2,660/64,349 (4.1%)	2,506/61,537 (4.1%)	154/2,812 (5.5%)	

Table 1 continued

Factor	Total <i>N</i> = 64,349	No AKIN = 61,537	AKIN = 2,812	<i>P</i> value
Other	616/64,349 (1.0%)	598/61,537 (1.0%)	18/2,812 (0.6%)	
Surgery duration (min), median [IQR]	215 [163–290]	214 [162–287]	254 [188–353]	< 0.001 ^c
Intraoperative TWA < 65 mm Hg, median [IQR]	0.06 [0.00–0.30]	0.06 [0.00–0.30]	0.08 [0.00–0.33]	< 0.001 ^c
Blood loss (mL), median [IQR]	100 [30–200]	100 [25–200]	200 [50–400]	< 0.001 ^c
Colloid administration (mL), median [IQR]	0 [0–500]	0 [0–500]	0 [0–500]	< 0.001 ^c

Data are presented as mean (SD), median [IQR], or *n*/total *N* (%)

^aANOVA

^bPearson's Chi square test

^cKruskal–Wallis test

Acetylsalicylic acid, nonsteroidal anti-inflammatory agents, and nephrotoxic antineoplastic agents were excluded from the confounder list because there were too few events (*N* < 5) by AKI

ACEI = angiotensin-converting enzyme inhibitors; AKI = acute kidney injury; ASA = American Society of Anesthesiologists; IQR = interquartile range; MAP = mean arterial pressure; POA = present on admission; PPIs = proton-pump inhibitors; SD = standard deviation; TWA = time-weighted average

study population (Fig. 1). In that population, 2,812 (4.4%) patients had postoperative AKI. Baseline and demographic characteristics are summarized in Table 1 by AKI status. All patients had at least one PACU measurement. Taking all PACU measurements as one averaged postoperative measurement and pooling it with postoperative ward measurements, the median [interquartile range (IQR)] number of postoperative MAP measurements was 17 [11–27]. The median [IQR] PACU duration was 176 [129–245] min. The median [IQR] number of creatinine measurements was 3 [1–4] during the first 7 postoperative days.

Primary analysis

The statistical analysis plan was included in our application and approved by the IRB. The original plan included an analysis of the association between the highest MAP and later AKI probability to assess the hypertensive harm threshold above which AKI risk increased. Nevertheless, only about 20% of our population experienced hypertension, defined as the highest MAP > 107 mm Hg,¹¹ which would provide insufficient power to analyze the hypertensive threshold. We therefore restricted our analysis to the lowest MAP to assess the hypotensive harm threshold, an aspect of the original statistical analysis plan that was not substantively modified after IRB submission.

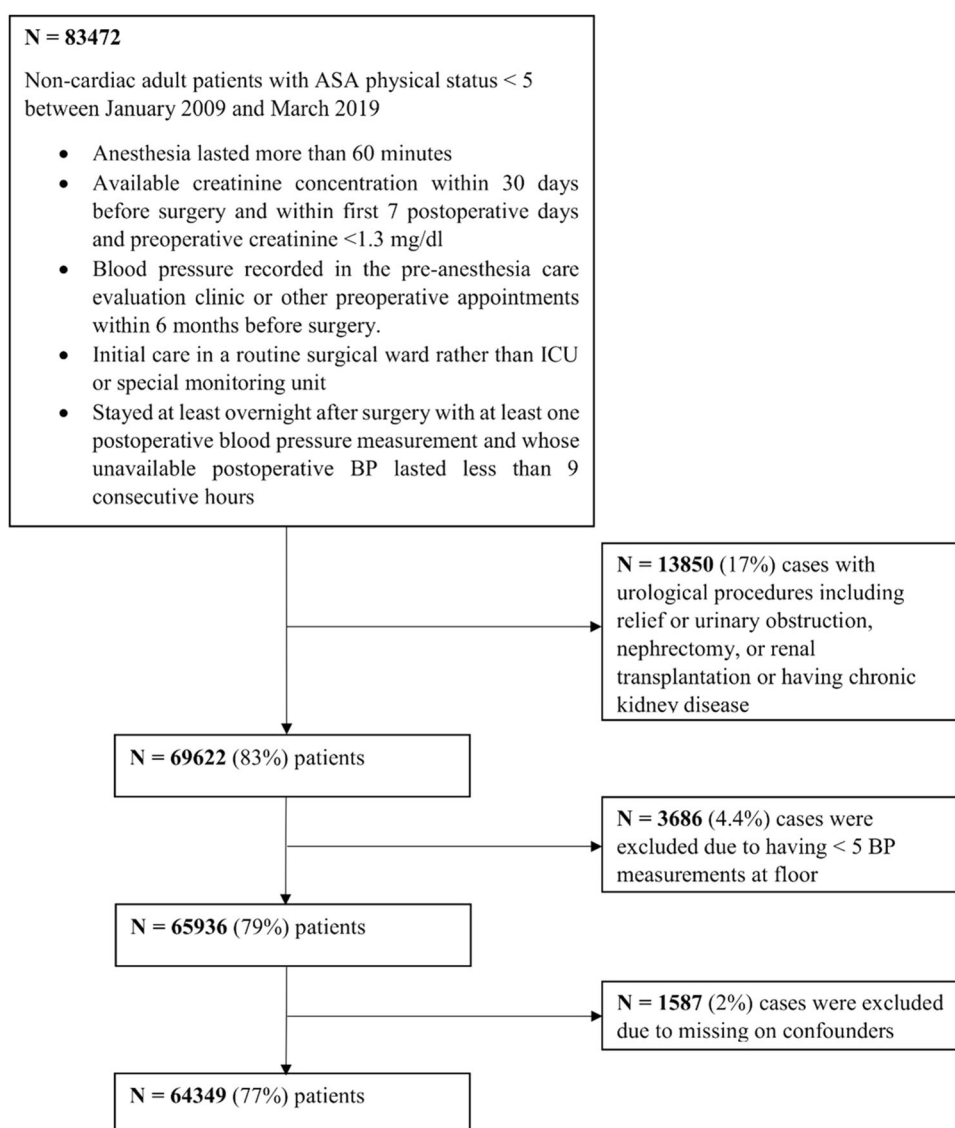
There were interactions between baseline blood pressure quartiles and the lowest MAP (*P* < 0.001), but as shown in the ESM eFigure, the interaction was only quantitative, not qualitative (i.e., same direction but apparently with different slopes). We therefore chose to ignore the interactions with baseline blood pressure and focused on the marginal relationship between the lowest MAP and AKI.

The relationship between the lowest MAP and AKI was nonlinear (Fig. 2). The odds ratio (OR) of AKI associated with the lowest MAP when the lowest MAP was < 80 mm Hg was different from that when the lowest MAP was ≥ 80 mm Hg (linear spline *P* < 0.001). When the lowest MAP was < 80 mm Hg, the OR of AKI associated with each 5-mm Hg decrease was 1.28 (97.5% confidence interval [CI], 1.23 to 1.32; *P* < 0.001). Nevertheless, the association was not significant (significance criterion was 0.025 after correcting for multiple testing) when the lowest MAP was ≥ 80 mm Hg, with an OR of 1.08 (97.5% CI, 0.99 to 1.17; *P* = 0.04) for each 5-mm Hg decrease in the lowest MAP (Table 2).

When categorizing the lowest MAP into low (≤ 60 mm Hg), medium (60–80 mm Hg), and high (> 80 mm Hg) groups, the odds of AKI was significantly higher for patients with the lowest MAP ≤ 60 mm Hg (OR, 3.22; 97.5% CI, 2.68 to 3.88; *P* < 0.001) or 60–80 mm Hg (OR, 1.63; 97.5% CI, 1.43 to 1.86; *P* < 0.001) compared with > 80 mm Hg (ESM eTable 3).

In a post hoc sensitivity analysis, we tried to fit a multinomial logistic regression model with all the stages being separated. Because there were too few events in stage 3 (*N* = 140), the model would not converge when we treated stage 3 separately. We therefore combined stages 2 and 3 together as stage 2/3. We provide the estimate of OR in ESM eTable 4. Stage 2/3 was more associated with MAP decrease when the MAP was < 80 mm Hg with OR > 1 for either of stage 2/3 vs stage 0 (OR, 1.49; 97.5% CI, 1.39 to 1.60) or stage 2/3 vs stage 1 (OR, 1.21; 97.5% CI, 1.12 to 1.31) with each 5-mm Hg decrease in lowest MAP. The OR of stage 1 vs stage 0 was also associated with the lowest MAP when the MAP was < 80 mm Hg (OR, 1.23; 97.5% CI, 1.19 to 1.28) with each 5-mm Hg

Fig. 1 Enrolment flow chart
ASA = American Society of Anesthesiologists; BP = blood pressure; ICU = intensive care unit



decrease in lowest MAP. When the lowest MAP was ≥ 80 mm Hg, the OR of any higher AKI stages *vs* lower stages was not significant.

Discussion

In this single-centre historical cohort analysis of 64,349 adults who had noncardiac surgery and had creatinine recorded preoperatively and postoperatively, the association between the lowest MAP and AKI was not significant when the lowest MAP exceeded 80 mm Hg. In contrast, the OR for AKI was significantly associated with each 5-mm Hg decrease when the lowest MAP was < 80 mm Hg, suggesting that a MAP of 80 mm Hg might be a harm threshold for AKI. We also found a quantitative, but not qualitative, interaction between baseline MAP and the relationship between the lowest postoperative MAP

component and later AKI incidence (e.g., associations in the same direction with various baseline pressures, but by different amounts). The postoperative MAP hypotensive threshold of 80 mm Hg thus applies over a broad range of preoperative blood pressures.

Our primary outcome variable was the new onset of AKI, that is, the occurrence of at least stage I AKI after noncardiac surgery. Stage I AKI, a seemingly mild level of renal dysfunction, is nonetheless clinically meaningful. For example, patients with stage I AKI after noncardiac surgery have 2.4 times higher odds of having long-term renal dysfunction than patients without postoperative kidney injury. Furthermore, a quarter of patients with postoperative stage I AKI remain in stage I AKI 1–2 years after the events, and an additional 11% had even worse renal function.⁴ The relationship between postoperative hypotension and AKI is thus clinically meaningful.

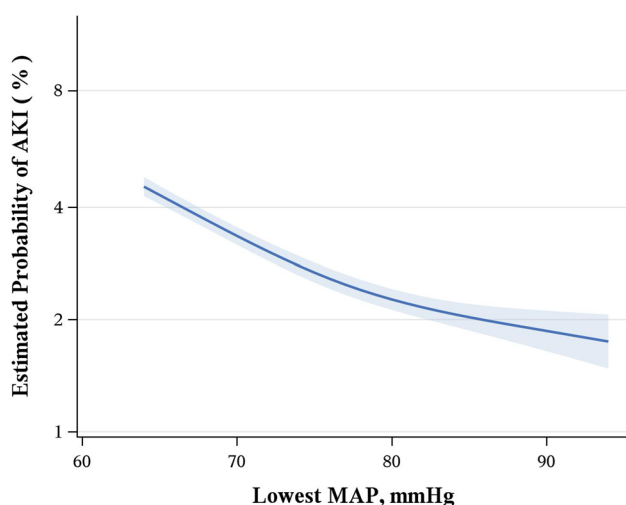


Fig. 2 The overall relationship between the lowest postoperative mean arterial pressure and acute kidney injury. The purpose of this plot was to visualize and explore the overall nonlinear relationship between the lowest postoperative MAP and AKI. Based on this plot, a piece-wise linear knot at 80 mm Hg for the lowest MAP was added at corresponding logistic regression model. The blue line represents the predicted probability of AKI and the blue band represents the 95% CI. AKI = acute kidney injury; MAP = mean arterial pressure; CI = confidence interval

Compared with other perfusion-dependent organs such as the heart and brain, the kidneys are especially susceptible to low perfusion pressure in nonsurgical settings, possibly because autoregulation is less effective. The effective range for pressure autoregulation¹² depends on pre-existing hypertension,^{13,14} diabetes mellitus,¹⁵ and various other factors. But very roughly, the lower perfusion pressure thresholds are thought to be about 60 mm Hg for the brain,¹⁶ 40 mm Hg for coronary arteries,¹³ and 80 mm Hg for kidneys.^{17,18} The MAP harm threshold for AKI thus appears to be higher than that for other perfusion-related organs.

Our results suggest that postoperative hypotension may contribute to AKI. Other results suggest that postoperative hypotension contributes to myocardial injury^{6–8} and other complications.¹⁹ Nonetheless, postoperative vital signs are usually measured intermittently at four-to-eight hour intervals. Perhaps consequently, hemodynamic abnormalities including hypotension are often unrecognized in patients recovering from surgery.⁵ It seems likely that more frequent—preferably continuous—monitoring will identify hypotension earlier than intermittent assessments, thus providing clinicians with opportunities to intervene and possibly reduce renal injury.

The most serious limitation of our analysis is uncertainty about when AKI developed because in most patients, creatinine was measured once daily or even less often. Furthermore, creatinine takes some time to increase after renal function degrades. A similar problem arises whenever exposure and outcomes overlap in time. We therefore limited our analysis to blood pressures that were recorded before AKI was identified. Fully addressing this issue will require a randomized trial in which patients were assigned to tight vs routine blood pressure management and creatinine is measured at least several times a day.

Another limitation of our analysis is that hypotension exposure was characterized by the average of the three lowest blood pressures, usually measured at four-hour intervals. Selection of three values was a trade-off between having sufficient data to average to reduce the influence of artifacts and having a reasonably short period. We used a single average MAP value during the PACU stay because the average PACU stay was only 176 min, which was shorter than the normal four-hour ward interval. Using just the lowest PACU values would have overweighted short periods of PACU hypotension relative to the ward exposure.

An additional limitation is that we only evaluated renal injury. Myocardial injury would have been of considerable interest, but we had too few patients with scheduled

Table 2 Primary analysis-unadjusted and adjusted relationship between the absolute change of postoperative mean arterial pressure and acute kidney injury

Exposure	Estimated OR, unadjusted (97.5% CI) ^a	<i>P</i> value	Estimated OR, adjusted (97.5% CI) ^b	<i>P</i> value
<i>Postoperative lowest MAP</i>	<i>5 mm Hg decrease in lowest MAP^c</i>			
< 80 mm Hg	1.29 (1.25 to 1.33)	< 0.001	1.28 (1.23 to 1.32)	< 0.001
≥ 80 mm Hg	1.04 (0.97 to 1.13)	0.22	1.08 (0.994 to 1.17)	0.04

^aThe estimates and *P* value were obtained from a univariate logistic regression model with lowest MAP being exposure and linear piece-wise knot at 80 mm Hg of lowest MAP. The *P* value was obtained from the Wald test

^bThe estimates and *P* value were obtained from a multivariable logistic regression model with lowest MAP being exposure and linear piece-wise knot at 80 mm Hg of lowest MAP adjusting for confounders in Table 1. The *P* value was obtained from the Wald test

^cWe adjusted significance level for relationship estimates below and above threshold within exposure using Bonferroni correction

AKI = acute kidney injury; CI = confidence interval; MAP = mean arterial pressure; OR = odds ratio

(not for-cause) troponin assessments. As with any observational analysis, unobserved confounding might exist despite a broad range of potential confounding factors adjusted in this study. And finally, while we show a potentially clinically meaningful association between hypotension and AKI, a trial will be required to determine the extent to which the association is causal, and thus amenable to intervention. We also note that, unlike the intraoperative period, it is difficult to effectively intervene for ward hypotension since vasopressors cannot be easily used.

Conclusions

In this historical cohort study, postoperative hypotension, defined as the lowest postoperative MAP < 80 mm Hg, was associated with AKI after noncardiac surgery. A prospective trial will be required to determine whether the observed association is causal and thus amenable to modification.

Author contributions Tetsuya Shimada helped with manuscript writing, study design, and data interpretation. Xuan Pu helped with programming, data acquisition, statistical analyses, and manuscript editing. Esra Kutlu Yalcin, Barak Cohen, and Mauro Bravo helped with data interpretation and manuscript editing. Edward J. Mascha helped as methodology advisor, and with statistical analyses, manuscript editing, and data validation. Alparslan Turan helped with study design, data interpretation, and manuscript editing. Daniel I. Sessler helped with study design, data interpretation, and manuscript editing.

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